

THE RELATION BETWEEN THE DEGREE OF RENAL FUNCTIONAL IMPAIRMENT AND HAEMOGLOBIN A₁ IN NONDIABETIC PATIENTS WITH CHRONIC RENAL FAILURE

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ABSTRACT:

The study was undertaken to investigate whether there is some correlation of the degree of renal functional impairment (as determined by serum creatinine and creatinine clearance rate) with haemoglobin A₁ in end stage renal disease. Thirty six nondiabetic patients were selected for the study. Four age groups and three disease groups of the patients were recognized. A weak inverse correlation was observed between the level of serum creatinine and haemoglobin A₁ in patients of age group 46 to 65 years ($r = -0.46$, $p < 0.1$).

INTRODUCTION

Haemoglobin A₁ (Hb A₁) is the minor component of normal adult haemoglobin. This haemoglobin is known as glycohaemoglobin. It is made up of subgroups A-E (Haemoglobin A_{1A}-HbA_{1E}). Although present at birth, the presence of Hb A₁ was not known until 1958 (Allen et al., 1958). This haemoglobin was first noted in 1958, but it was not until 1968 that extensive studies on HbA₁ were begun by Rahbar (Rahber, 1968). It was first thought to be an abnormal haemoglobin in diabetics; thus its original name of "diabetic haemoglobin". HbA₁ has now been recognized to be a minor haemoglobin component present in all people.

Bunn, 1981; Heinrichs et al., 1981 and Winterhalter, 1979 investigated that HbA₁ was composed of several fractions but haemoglobin A_{1C} (HbA_{1C}) was the largest fraction (about 70%) haemoglobin A (HbA) and being the actual glycosylated haemoglobin. HbA_{1C} has glucose molecule attached to the beta chain and the degree of glycosylation of haemoglobin A (HbA) is directly depend upon the concentration of blood glucose. HbA₁ and HbA_{1C} levels have been found to be two or three times greater in diabetics than in nondiabetics. It has been determined that HbA₁ and HbA_{1C} levels reduced to normal values when the blood glucose is well controlled. Since glycohaemoglobin levels represent a time averaged blood glucose levels, it is

considered to be a better control than individual blood glucose levels.

HbA₁ and HbA_{1c} have been found to be elevated in patients with chronic renal failure (CRF), despite the shortened life span of erythrocytes. This might be due to the fact that end stage of renal disease can result in disturbance of carbohydrates metabolism even in nondiabetic patients. A significant positive interdependence was observed between HbA₁ and fasting serum glucose levels in nondiabetic patients with CRF (Ahmed et al., 1990). Elevated HbA₁ levels were found in patients with CRF and in diabetic and nondiabetic patients undergoing dialysis. Increased HbA₁ levels in CRF may be caused by an increase in glycosylated haemoglobin secondary to the associated glucose intolerance (De Boer et al., 1981 and Scherthaner et al., 1979).

The suggestion was also made that the increased levels of those fast-movement chromatographic fractions may be caused either by increased glycosylated haemoglobin and or other haemoglobin species which were modified by yet unidentified reactant eluting with glycosylated haemoglobin component on cation exchange chromatography (De Marchi et al., 1981). The increased HbA₁ in CRF was largely caused by haemoglobin carbamylation resulting from a condensation of urea-derived cyanate with N-terminal groups. It was also determined that the extent of haemoglobin carbamylation by measuring the valine hydantoin released from carbamylated N-termini of the alpha chains and beta chains of haemoglobin upon acid hydrolysis, and found it to be proportional to the time averaged concentration of blood urea (Flukinger et al., 1981). Scherthaner et al. (1979), found that haemoglobin A₁ levels did not differ between patients with moderately impaired renal function and control but patients with severe renal insufficiency the haemoglobin A₁ levels were significantly increased. They showed that the increase in haemoglobin A₁ was significantly correlated with the plasma creatinine.

Purpose of Study

We were interested to see whether there is some correlation between degree of renal functional impairment (as determined by creatinine and creatinine clearance) and haemoglobin A₁ in patients with chronic renal failure.

MATERIALS AND METHODS

Thirty six patients of chronic renal failure were studied. The patients with diabetes and who received blood within one month before proceeding the study were not included. Patients selected were outdoor and indoor patients at the Department of Nephro- Urology, Jinnah Postgraduate Medical Centre, Karachi, Pakistan. Patients were classified on the basis of age into four groups ranging from 6 to 25, 26 to 45, 46 to 65 and 6 to 65 years. Three major disease groups were also made included hyperten-

sive chronic renal failure, nephrotic chronic renal failure and chronic renal failure due to chronic nephritis.

Blood sample was collected from each patient in the morning after an overnight fasting of 12 to 14 hours by an evacuated blood collecting plain glass tube (Terumo Corporation, Tokyo, Japan). The serum was either separated and stored at 4°C for assay within a week or frozen at -20°C if assay was to be further delayed. Sample taken in EDTA was stored but not for more than twelve days at +2 to 8°C in a refrigerator for the estimation of haemoglobin A₁. 24 hour's urine was collected during two consecutive days, for the evaluation of creatinine clearance.

The estimation of percentage of haemoglobin A₁ of total haemoglobin was done by the method of Schnek and Schroeder, 1961, (Boehringer Mannheim). Serum and urinary creatinine were measured by the modified method Owen et al. (1954) cited from Henry (1974).

RESULTS

The study was carried out in thirty six patients with chronic renal failure (CRF). Four age groups of the patients were recognized. Patients were also classified into three major disease groups on the basis of clinical examination, biochemical investigation, biopsy findings, intravenous pyelogram (IVP) and ultrasound reports. The results were summarized and mean + SEM of each group is shown in table 1, 2 and 3. HbA₁ levels of the patients were plotted against the values of serum creatinine (fig. 1-6) and creatinine clearance rate (fig. 7-12).

A weak significant negative correlation was observed between the level of serum creatinine and HbA₁ in patients with CRF of age group 46 to 65 years (fig. 3). There was no interrelationship found between the levels of serum creatinine and HbA₁ in patients with CRF of various age groups (fig. 1, 2 and 4) except age group 46 to 65 years. No significant interdependence was also observed between the above mentioned two parameters in patients with CRF due to hypertension and chronic nephritis (fig. 5 and 6). The values of correlation coefficient between serum creatinine and HbA₁ in patients with CRF were 0.225 ($P < 0.1$), -0.054 ($P < 0.1$), -0.46 ($P < 0.1$) and 0.117 ($P < 0.1$) for age groups 6 to 25, 26 to 45, 46 to 65 and 6 to 65 years respectively. The results of correlation coefficient between serum creatinine and HbA₁ were -0.204 ($P < 0.1$) and 0.39 ($P < 0.05$) for patients with CRF due to hypertension and chronic nephritis respectively.

A slight, but significant positive correlation between HbA₁ and creatinine clearance in patients with CRF of age groups 26 to 45 and 46 to 65 years (fig. 8 and 9)

was shown. There was direct relationship also observed between HbA₁ and creatinine clearance in patients with CRF due to hypertension (fig. 11). No interdependence was observed between the levels of HbA₁ and creatinine clearance in patients with CRF of age groups 6 to 25 and 6 to 65 years (fig. 7 and 10) as well as CRF patients due to chronic nephritis (fig. 12). The results of correlation coefficient between HbA₁ and creatinine clearance were -0.153 ($P < 0.1$), 0.447 ($P < 0.05$), 0.430 ($P < 0.1$) and 0.018 ($P < 0.1$) for age groups 6 to 25, 26 to 45, 46 to 65 and 6 to 65 years respectively. The values of correlation coefficients between HbA₁ and creatinine clearance 0.473 ($P < 0.05$) and -0.271 ($P < 0.1$) for patients with CRF due to hypertension and chronic nephritis.

Table 1
Age and sex distribution in normal healthy control subjects

Age Group (Years)	Male	Female	Age (Mean $\pm$ SEM)
6 - 25	8	2	16.70 $\pm$ 2.44
26 - 45	12	6	30.44 $\pm$ 1.24
46 - 65	4	2	52.00 $\pm$ 1.29
6 - 65	24	10	30.20 $\pm$ 2.26

Table 2
Age and sex distribution in patients with chronic renal failure

Age Group (Years)	Male	Female	Age (Mean $\pm$ SEM)
6 - 25	7	5	20.16 $\pm$ 5.82
26 - 45	11	4	38.84 $\pm$ 1.20
46 - 65	7	2	54.66 $\pm$ 2.29
6 - 65	25	11	36.55 $\pm$ 2.40

Table 3
Age and sex distribution in patients of various disease groups

Age Group (Years)	Male	Female	Age (Mean $\pm$ SEM)
CRF/Hypertension	10	5	37.66 $\pm$ 3.21
CRF/Chronic Nephritis	14	2	34.06 $\pm$ 3.83
CRF/Nephrotic Syndrome	3	2	33.80 $\pm$ 4.71

Values are expressed as Mean $\pm$ SEM.

DISCUSSION

The degree of renal functional impairment can be determined by measurement of glomerular filtration rate which is commonly approximated by the serum creatinine (Edwards and White, 1959; Effersoe, 1957; Enger and Blegen, 1964) and creatinine clearance rate. Serum creatinine concentration may vary in the presence of muscular wasting and decreased creatinine production. When clearance values are unavailable, serum creatinine concentration (secondary criterion) may be used. The council on the kidney in cardiovascular disease (1971), classify the severity of renal disorders on the basis of serum creatinine and creatinine clearance.

During present work we have estimated serum creatinine, urinary creatinine excretion and creatinine clearance rate in patients with CRF and control subjects. A significant increase in the levels of serum creatinine and decrease in creatinine clearance rate was observed in patients with CRF of various age groups as well as the disease groups when compared with normal control subjects. These observations are characteristics feature of patients with CRF.

Urinary creatinine excretion was found to be significantly low in patients with CRF of all disease and age groups except the patients of young age (6 to 25 years) groups perhaps either due to conservative treatment or moderately impaired renal function.

Caspaire and Miedema (1977) have diverted the attention to elevation of HbA₁ in renal failure. Increased level of HbA₁ levels have also been reported in patients with chronic renal haemodialysis. Similar observation made by Stanton et al. (1978) who

also reported that this increase may be due to coexisting hyperglycaemia. Further suggested that the increased levels of HbA₁ do emphasize the importance of controlling glucose intolerance in renal failure.

Oimomi et al. (1982) investigated HbA₁ levels in nondiabetic patients with CRF. They found a better positive correlation between HbA₁ and blood urea nitrogen than between HbA₁ and serum creatinine. Our patients with CRF of old age (46 to 65 years) group shows a weak negative correlation between HbA₁ and serum creatinine (fig. 3). These results contradicted the findings made by Oimomi et al. (1982). The decrease in HbA₁ with increase in serum creatinine in this age group shows that with the increase in the degree of renal functional impairment is probably related to the shortened life span of erythrocytes known to occur in patients with CRF.

Present data also shows that poor, but significant positive correlation between HbA₁ and creatinine clearance rate is present in patients with CRF of middle age (26 to 45) and old age (45 to 65) groups (fig. 8 and 9). Similar correlation also found in these two biochemical parameters in patients with CRF due to hypertension (fig. 11).

HbA₁ was also correlated with serum creatinine levels in patients with renal transplant recipient. There was no significant correlation between serum creatinine levels and HbA₁ concentration have been described in either the nondiabetic transplant recipients or for all renal transplant recipients (Zawada et al., 1985).

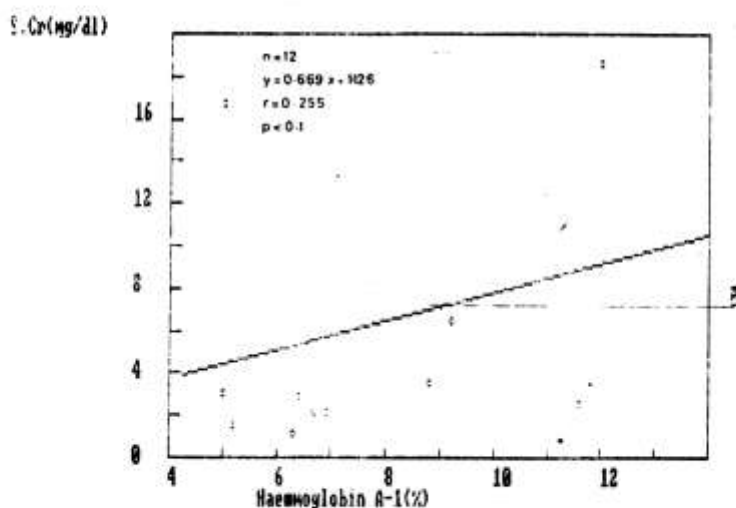


Fig. 1: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 6-25 years.

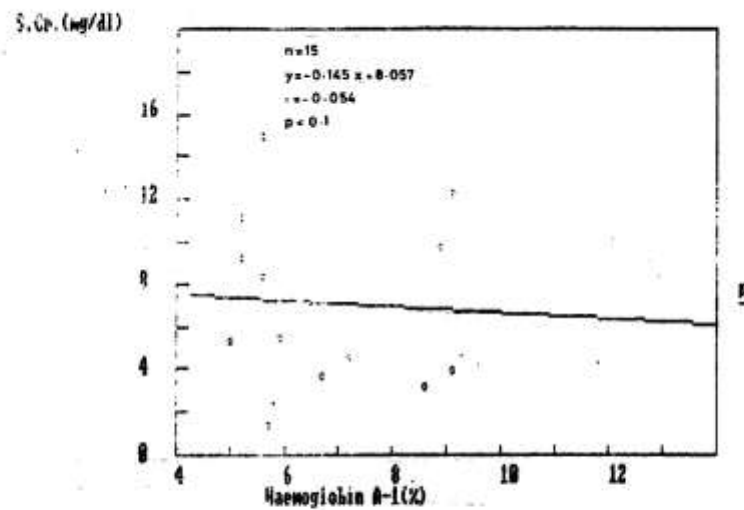


Fig. 2: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 26-45 years.

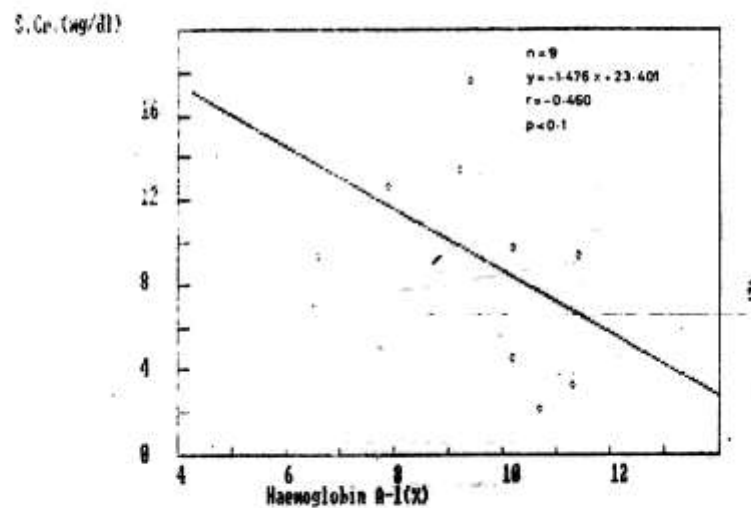


Fig. 3: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 46-65 years.

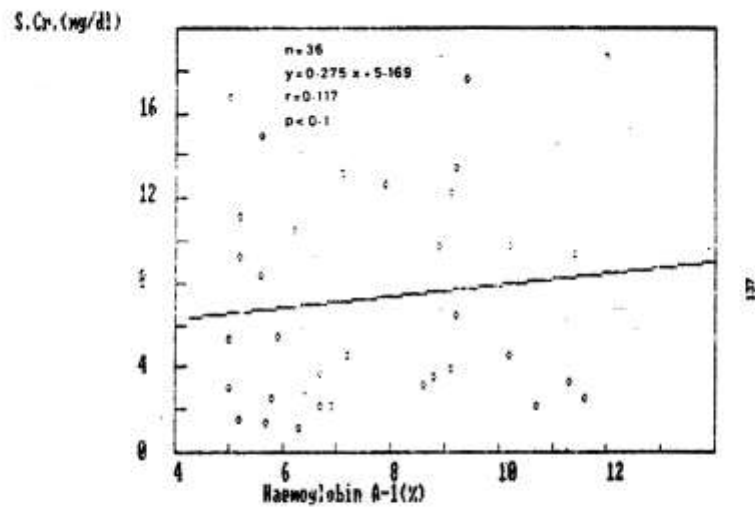


Fig. 4: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 6-65 years.

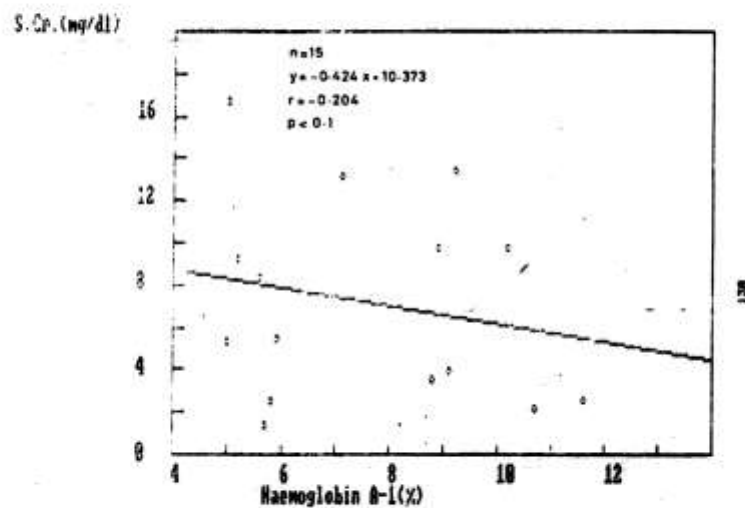


Fig. 5: Haemoglobin A-1 levels against serum creatinine in patients with CRF/Hypertension.

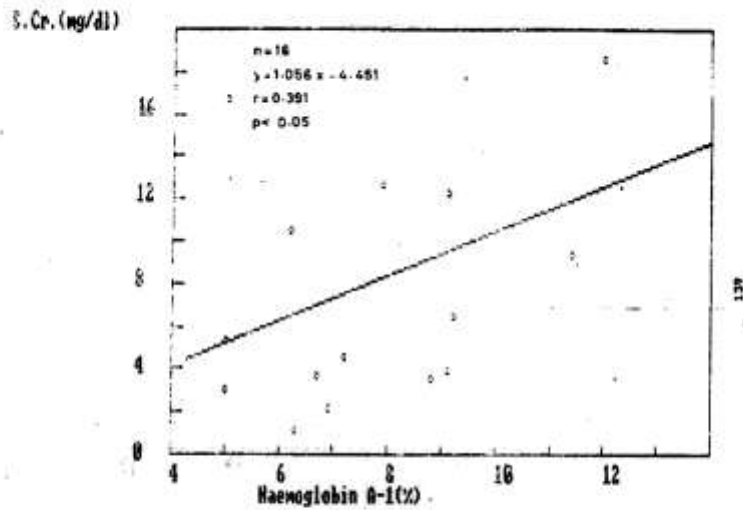


Fig. 6: Haemoglobin A-1 levels against serum creatinine in patients with CRF/Chronic Nephritis.

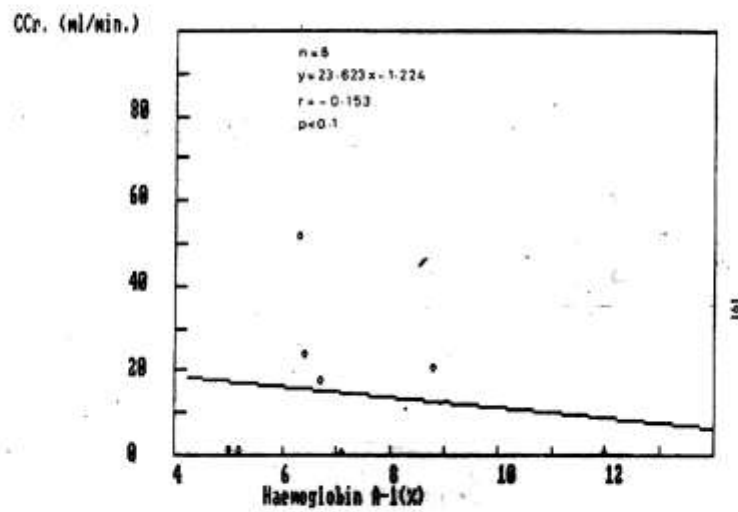


Fig. 7: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 6-25 years.

Relation between the degree of renal function

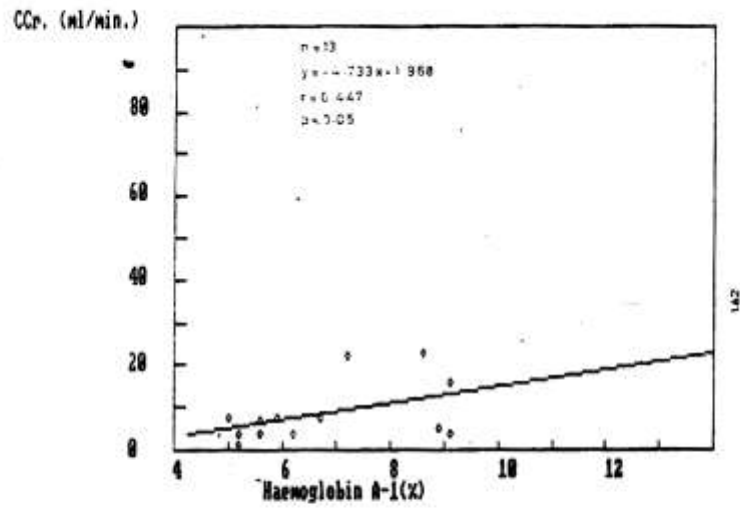


Fig. 8: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 26-45 years.

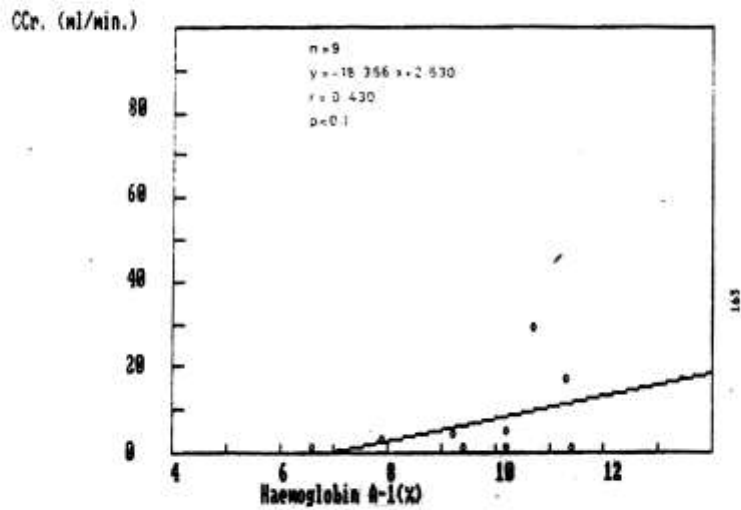


Fig. 9: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 46-65 years.

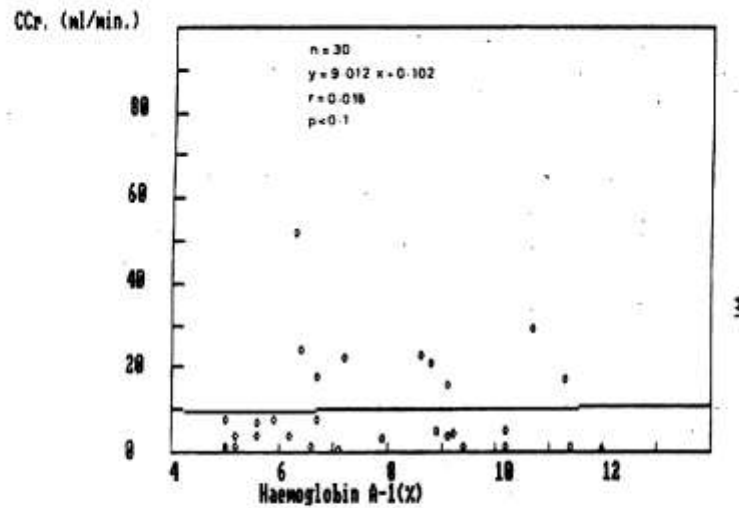


Fig. 10: Haemoglobin A-1 levels against serum creatinine in patients with CRF for age 6-65 years.

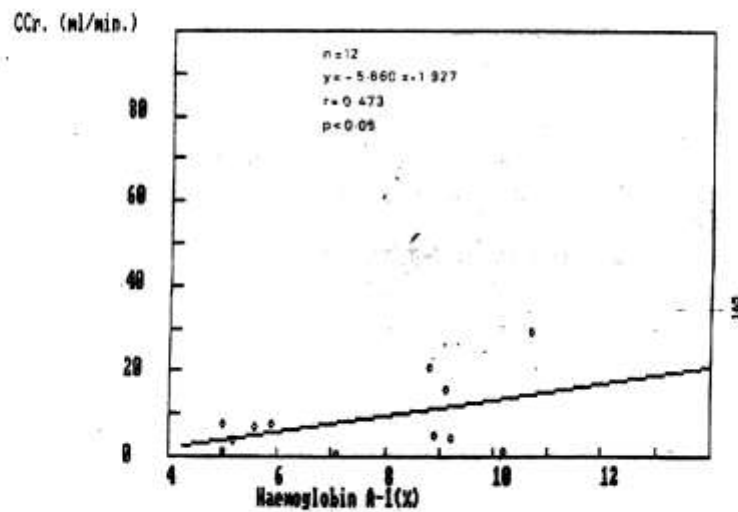


Fig. 11: Haemoglobin A-1 levels against serum creatinine in patients with CRF/Hypertension.

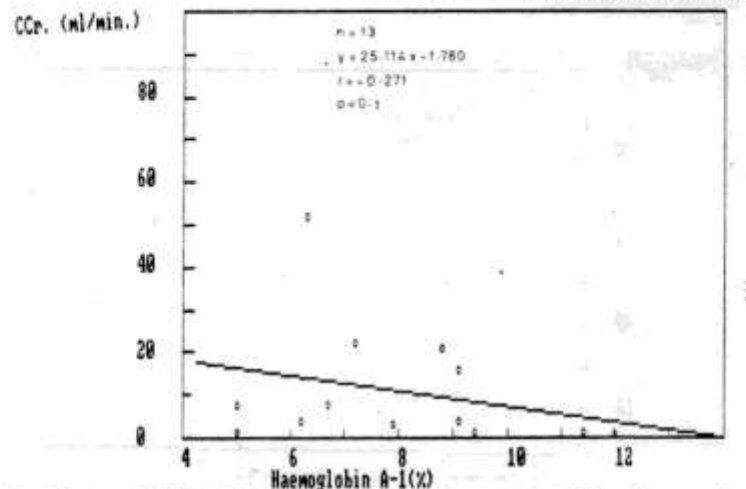


Fig. 12: Haemoglobin A-1 levels against serum creatinine in patients with CRF/Chronic Nephritis.

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